

Solr: The Platform that Makes Search Easy

This article gives a detailed description of Solr. It is aimed at geeks who have built a Web application and want to make the search for it faster.



Solr (pronounced Solar) was created by Yonik Seeley in 2004. Apache Solr is an open source search platform based on the Apache Lucene search library. In a sense, Solr is a cover over the Apache Lucene library. It uses the Lucene class to create an inverted index. Apache Solr is built upon Java. Most search websites are today built upon Solr, of which the most popular are e-commerce sites. Solr runs on standalone full-text search servers. The Lucene Java search library is used to create an index, which is used by Solr to perform searches. Solr has an HTML/XML interface and JSON APIs on which search applications are created. In layman's terms, Solr can be defined as follows: "Apache Solr is a search engine—you index a set of documents and then query Solr to return a set of documents that matches the user query."

Solr gained popularity because it is fast, and can index and search multiple sites. It returns relevant content based on the search query's taxonomy. Other important features are: faceted searching, dynamic clustering, database integration, NoSQL features, rich document handling, a comprehensive administration interface, high scalability and fault tolerance, as well as real-time indexing.

The preferred environment for the purpose of this article is Solr-5.3.0. Before you begin the Solr installation, make sure you have JDK installed and *Java_Home* is set appropriately.

Solr makes search easy

To understand how Solr makes search easy, we first have to configure it. As I am using Windows 7, I will demonstrate Solr in Windows. Solr is open source, so it can easily be downloaded from the Solr website. But remember, to run Solr, you need to have JRE installed in your system. For a Windows system, download the .zip file. To get started, all you need to do is extract the Solr distribution archive to a directory of your choosing. Once extracted, you are ready to run Solr in your system.

You can start Solr by running `bin\Solr.cmd`, if you are using Windows. This will start Solr on port 8983. If everything is fine, Solr will start by saying, 'Happy Search!'. If you want to change the port number, then give the command `Solr start -p port number`, using cmd. You can also use the `Solr -help` command to know the various types of commands in Solr. When Solr starts running, it will generate a URL to see the admin console. Commonly, the URL is `http://`